

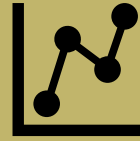
Introduction to Machine Learning (ML)

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Outline

- What is Machine Learning?
- ML Program Paradigm
- 3 Types of ML : Supervised, Unsupervised, Reinforcement Learning
- Basic Terminology
- Building ML system
- Edge v.s. Cloud Computing
- Python & Jupyter notebook guide

What is machine learning?



Algorithms that analyze data and turn into knowledge

Detect patterns/structures
Classification/regression
Predict future events

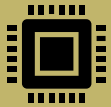


Subclass of AI (Artificial Intelligence)



Trained instead of programmed

Famous quotes

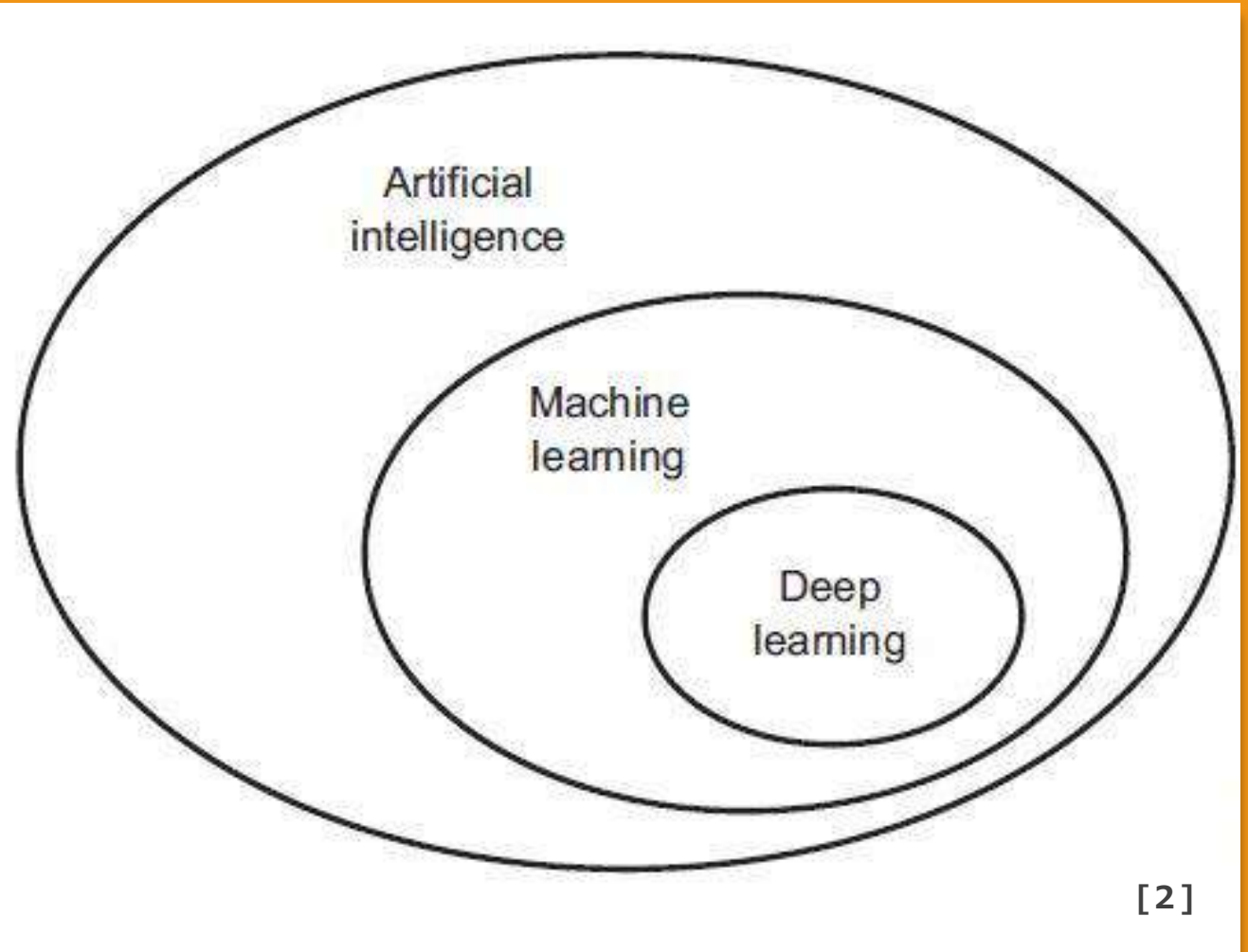


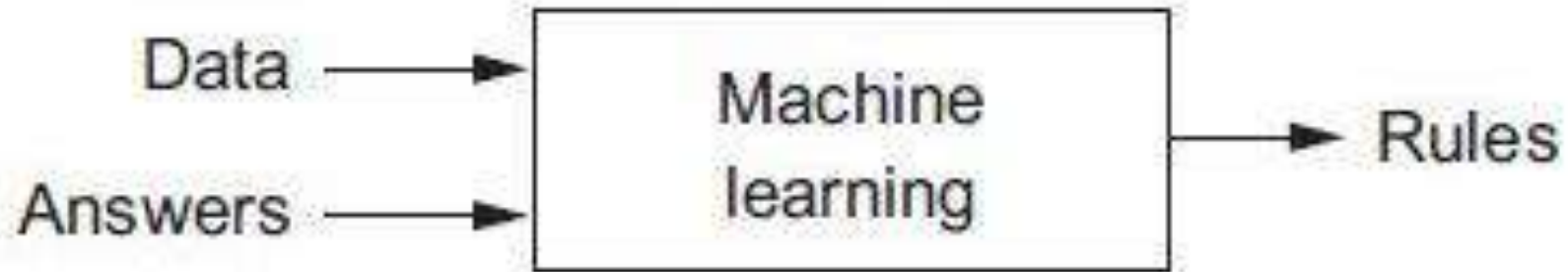
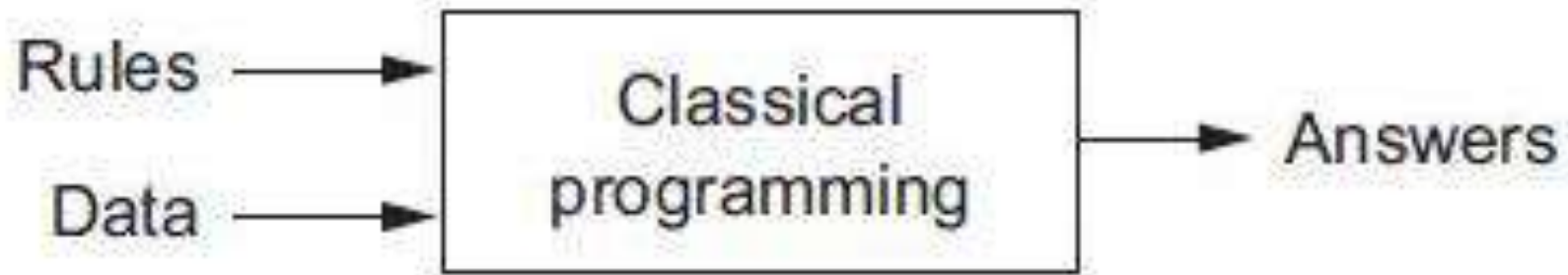
“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.” - Arthur L. Samuel . 1959.



“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” - Tom Mitchell, CMU. 1997.

AI/ML/DL





New program paradigm in ML

[2]

3 types of ML

Supervised

- labeled data (“ground truth”)
- direct feedback
- predict outcome/future

Unsupervised

- no labels/targets
- no feedback
- find hidden structure in data

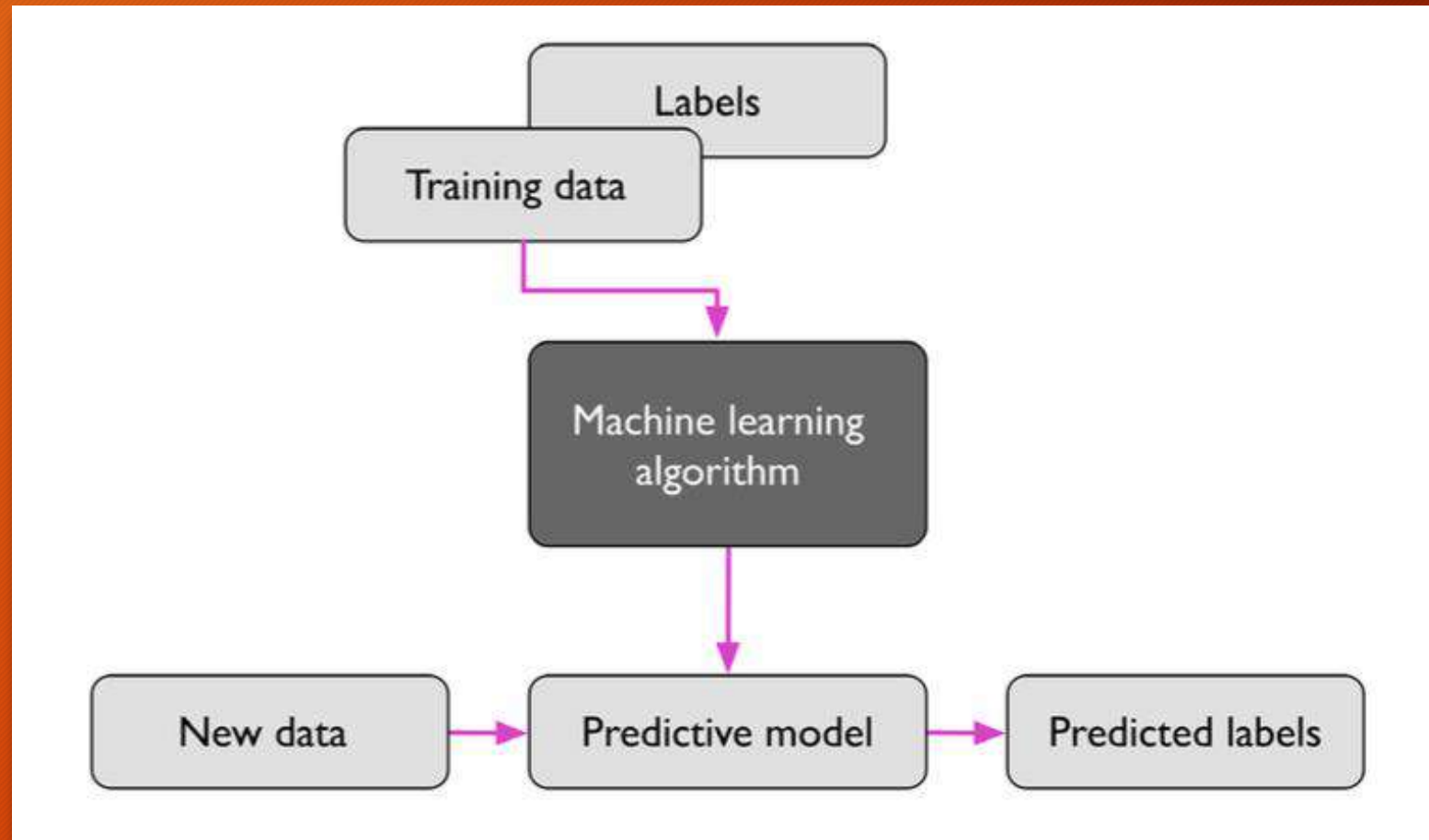
Reinforcement learning (RL)

- decision process
- reward system
- learn series of action

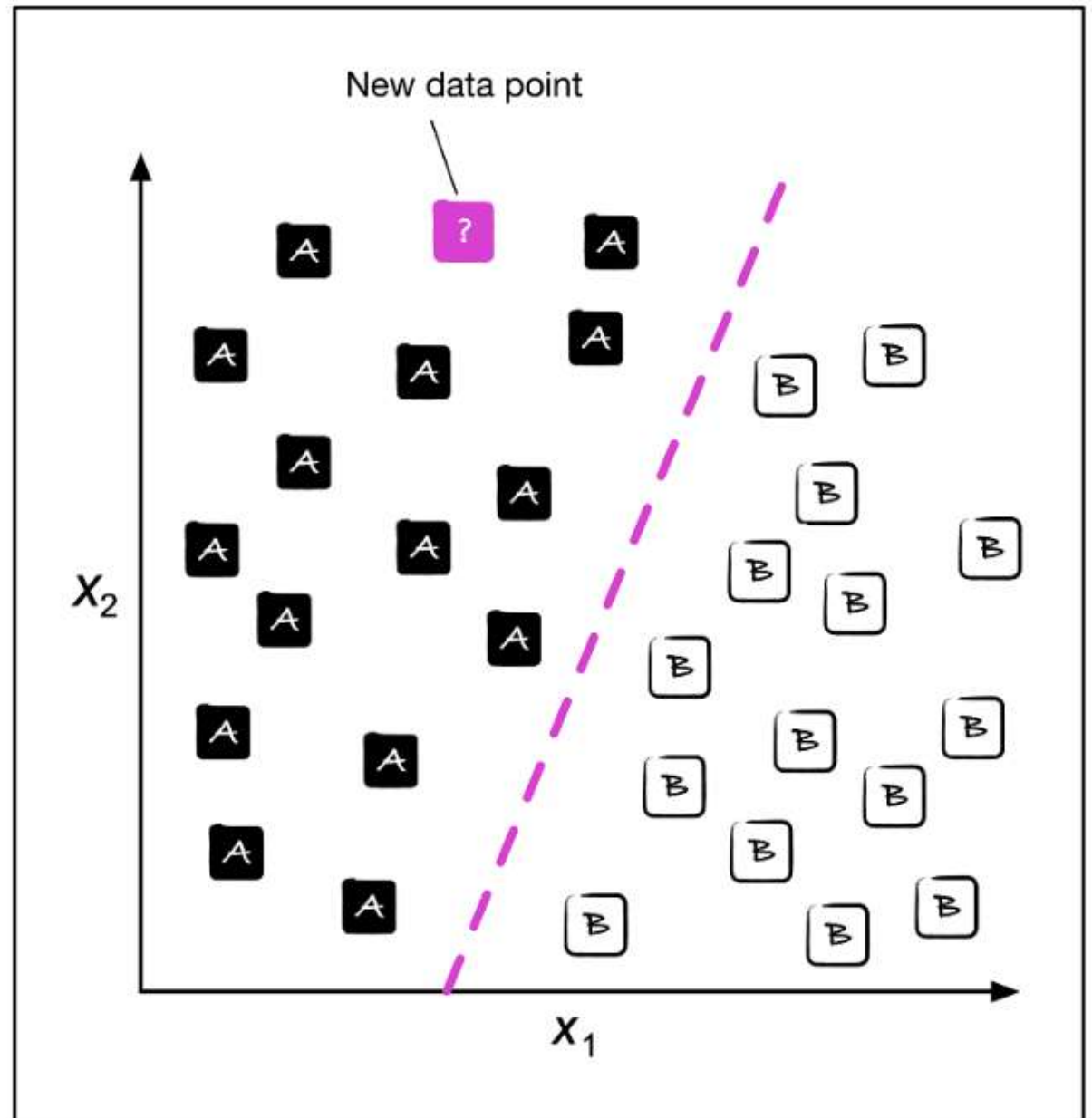
Supervised learning

Supervised learning process

- [1]



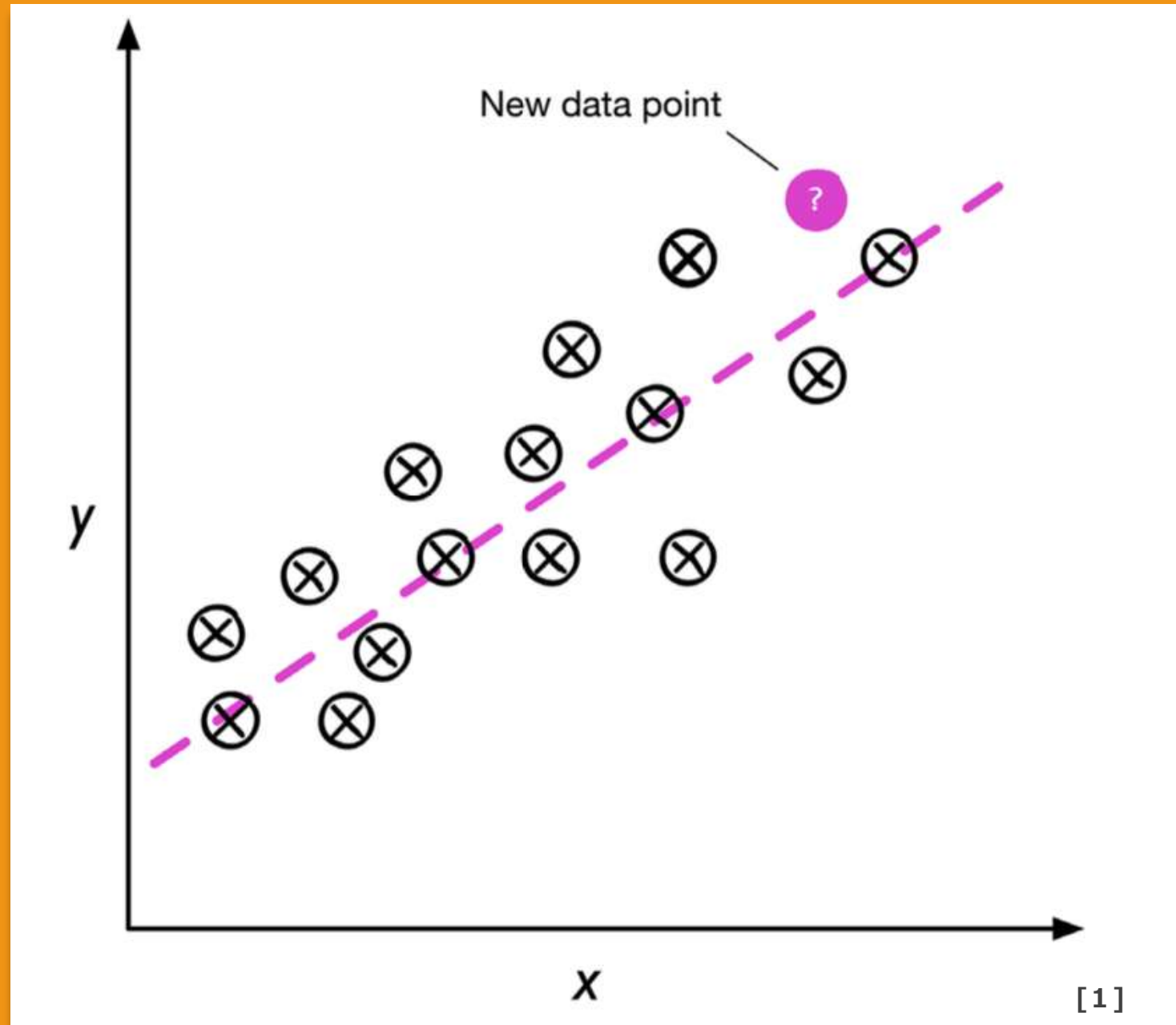
Classifying a new data point



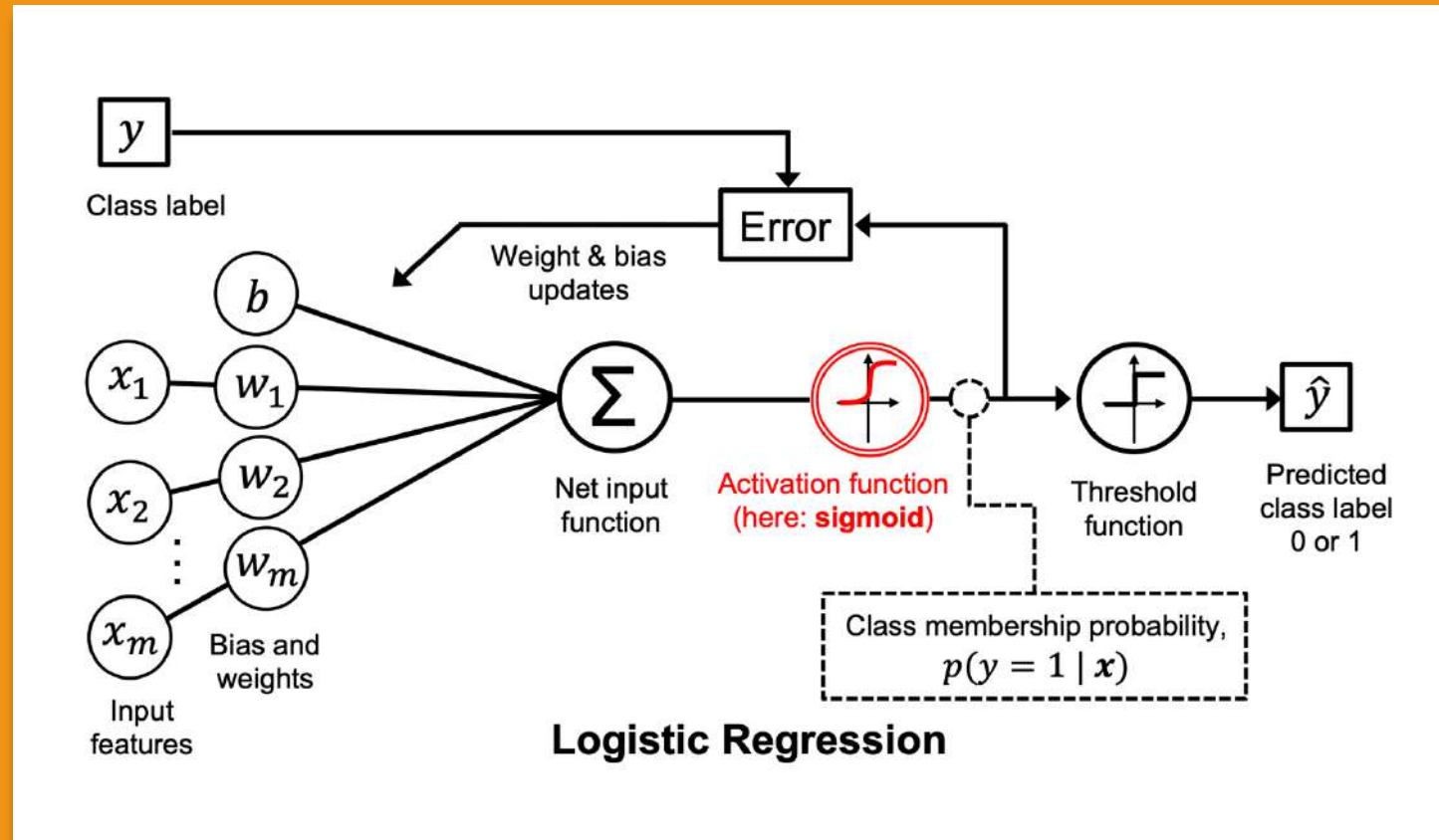
Examples of Supervised Learning

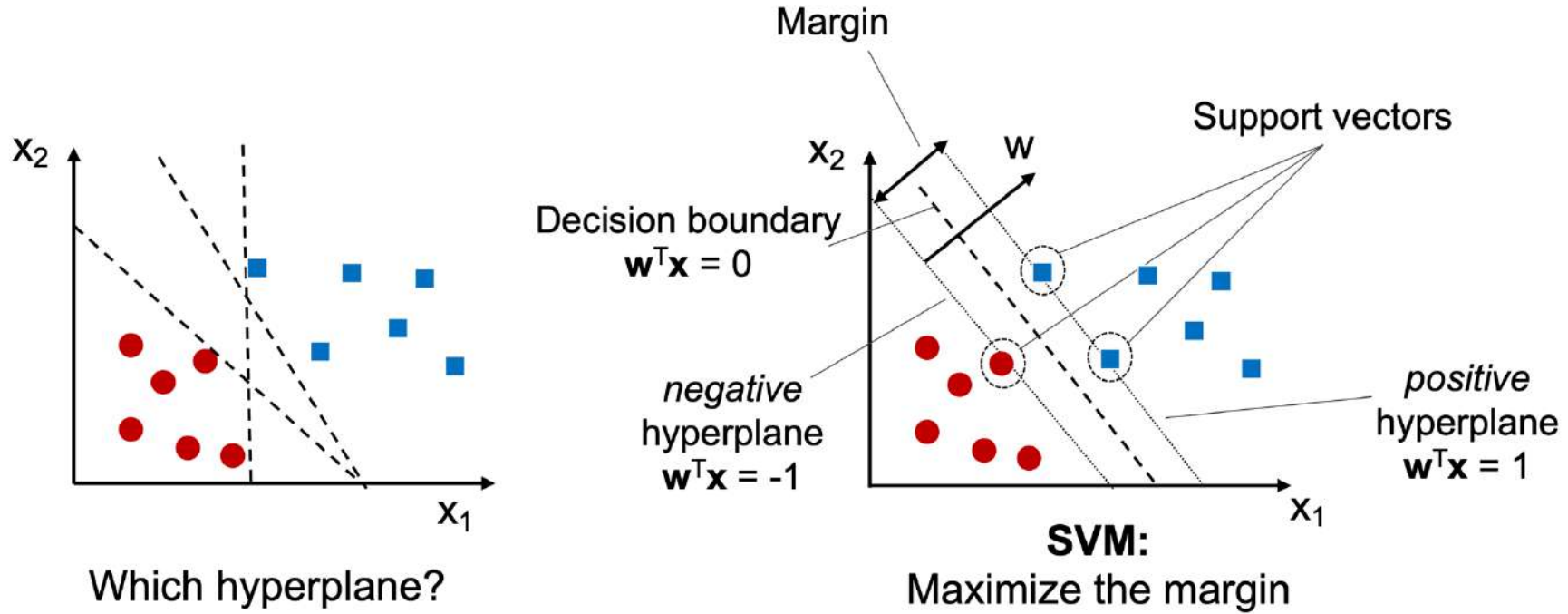
- Classical ML
 - Linear/Logistic Regression
 - Support Vector Machines
 - k-nearest neighbor
- Deep Learning
 - MLP/DNN
 - CNN

Linear regression



Logistic Regression

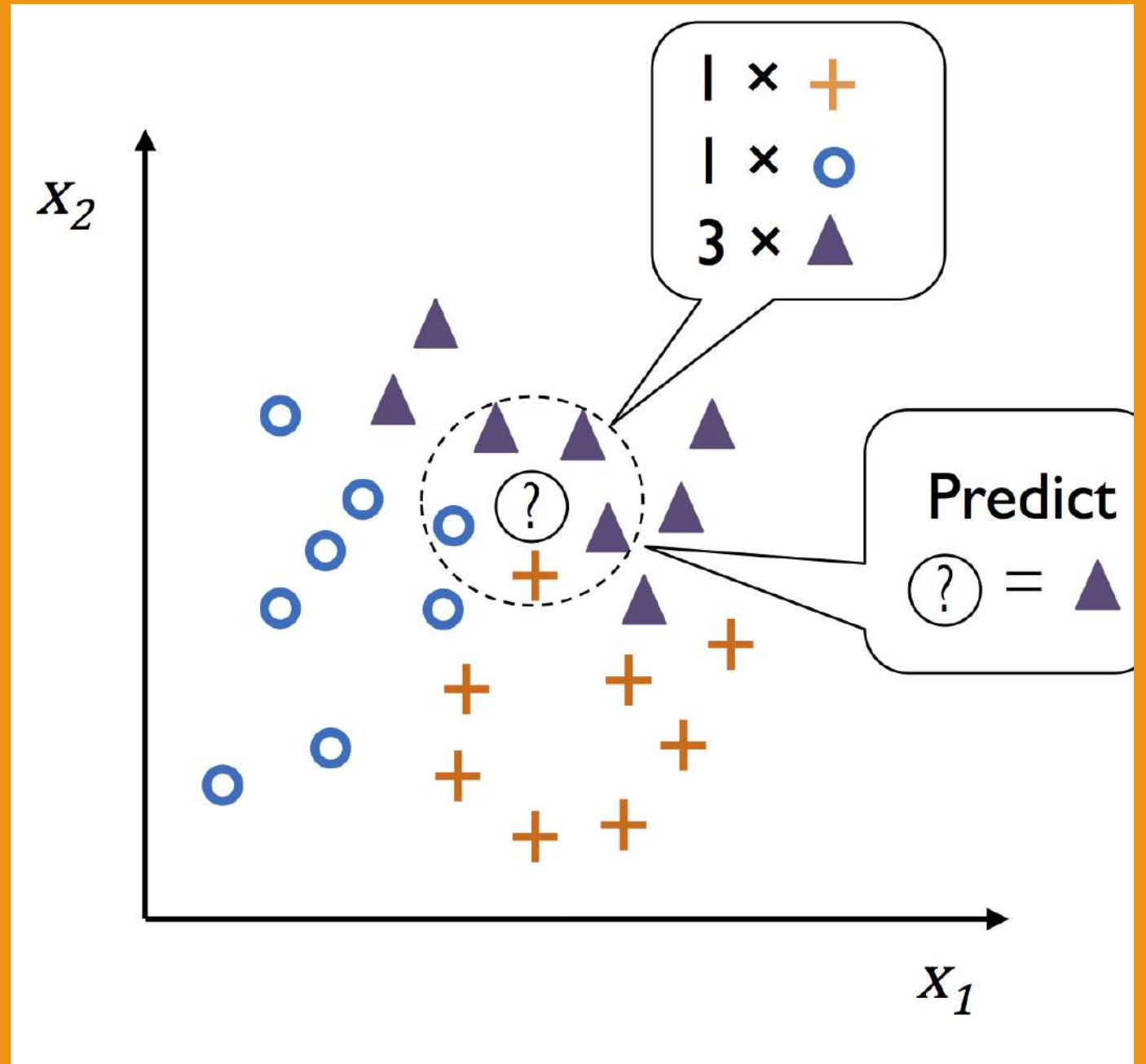




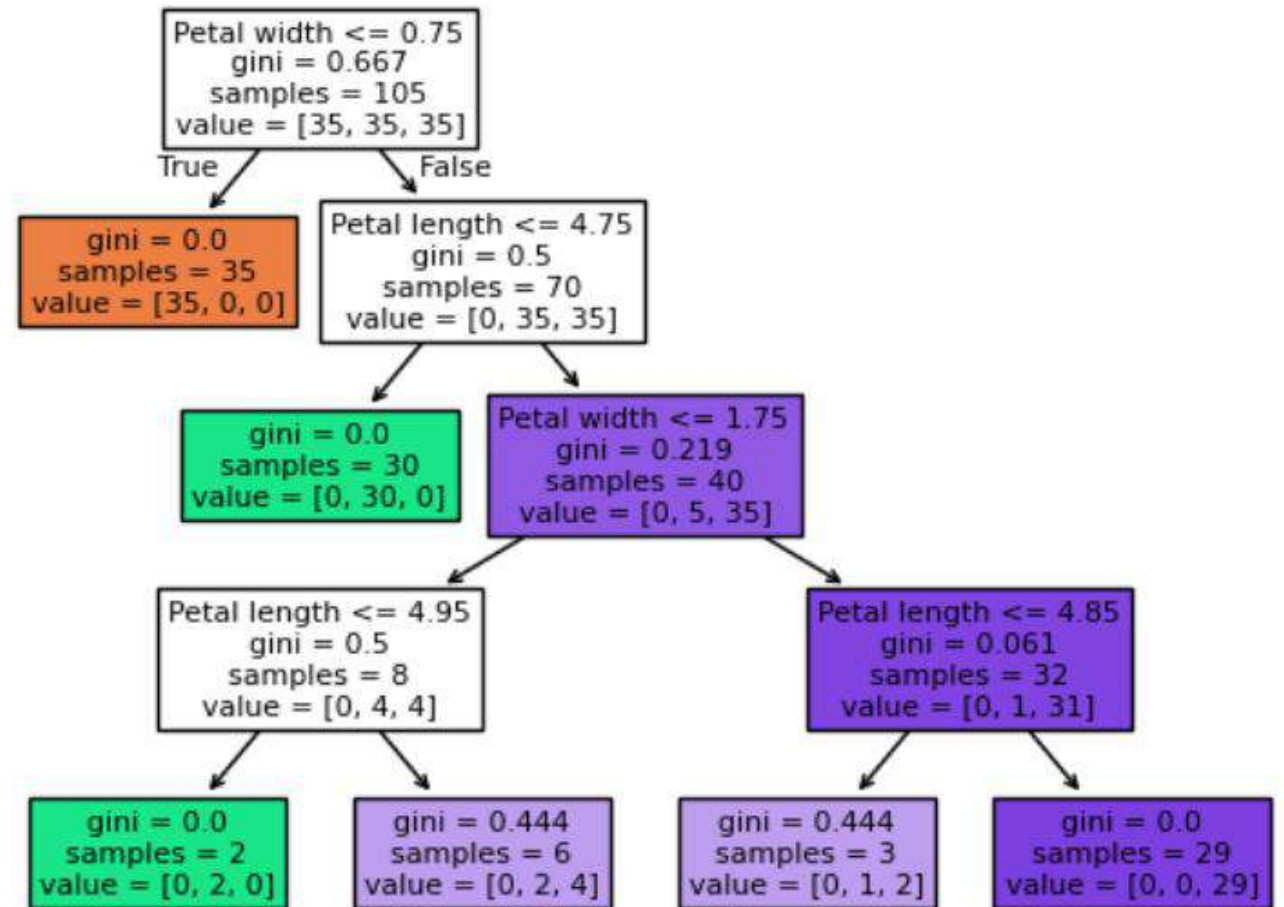
From Fig 3.10
of [1]

Support Vector Machines

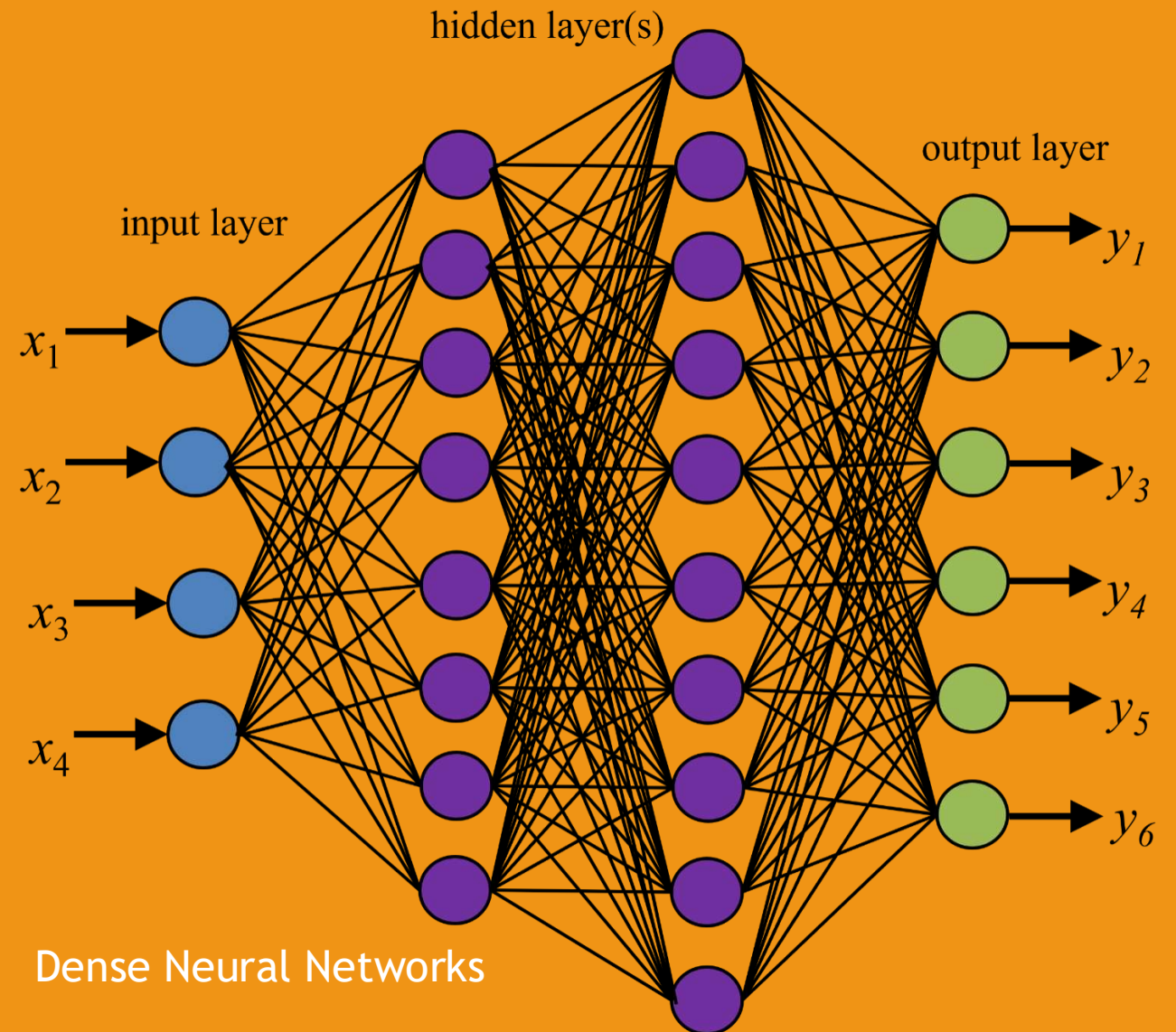
k-nearest neighbor



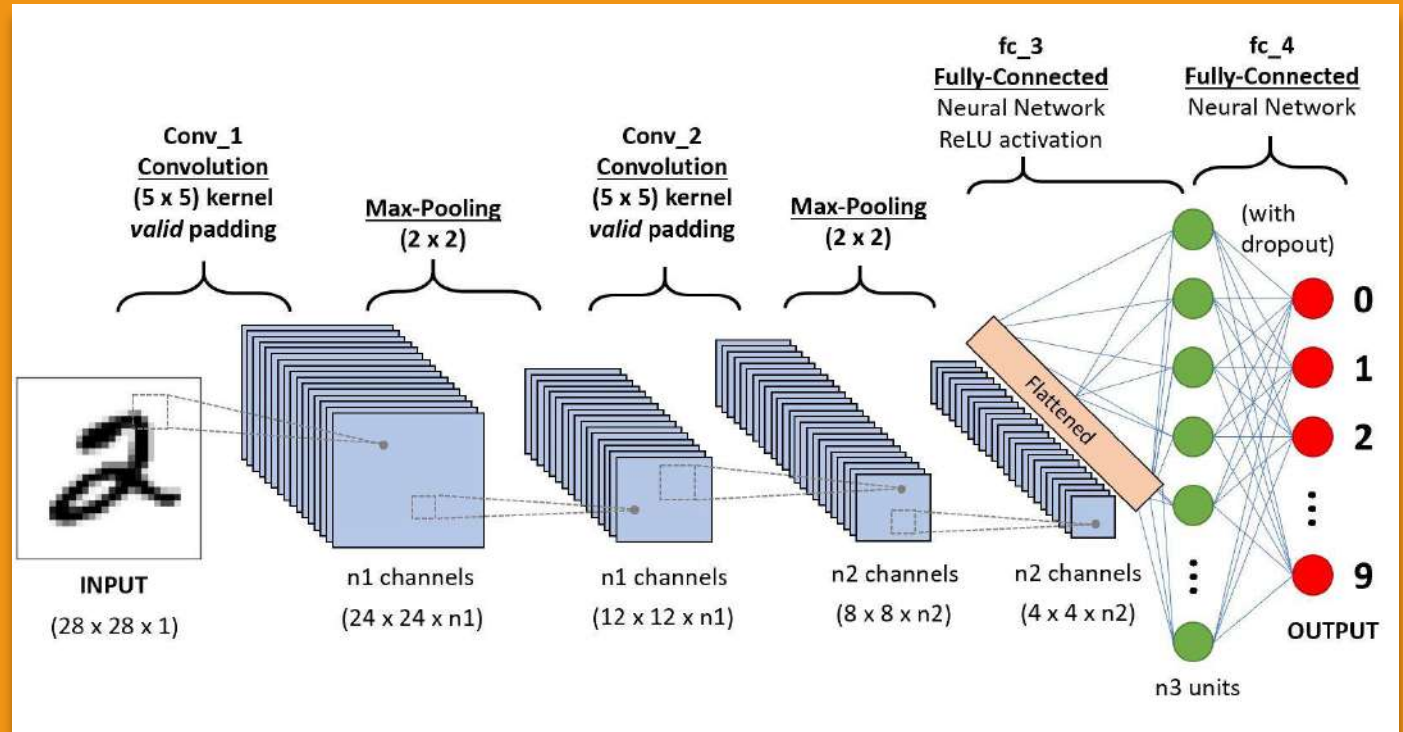
decision tree



Deep Learning



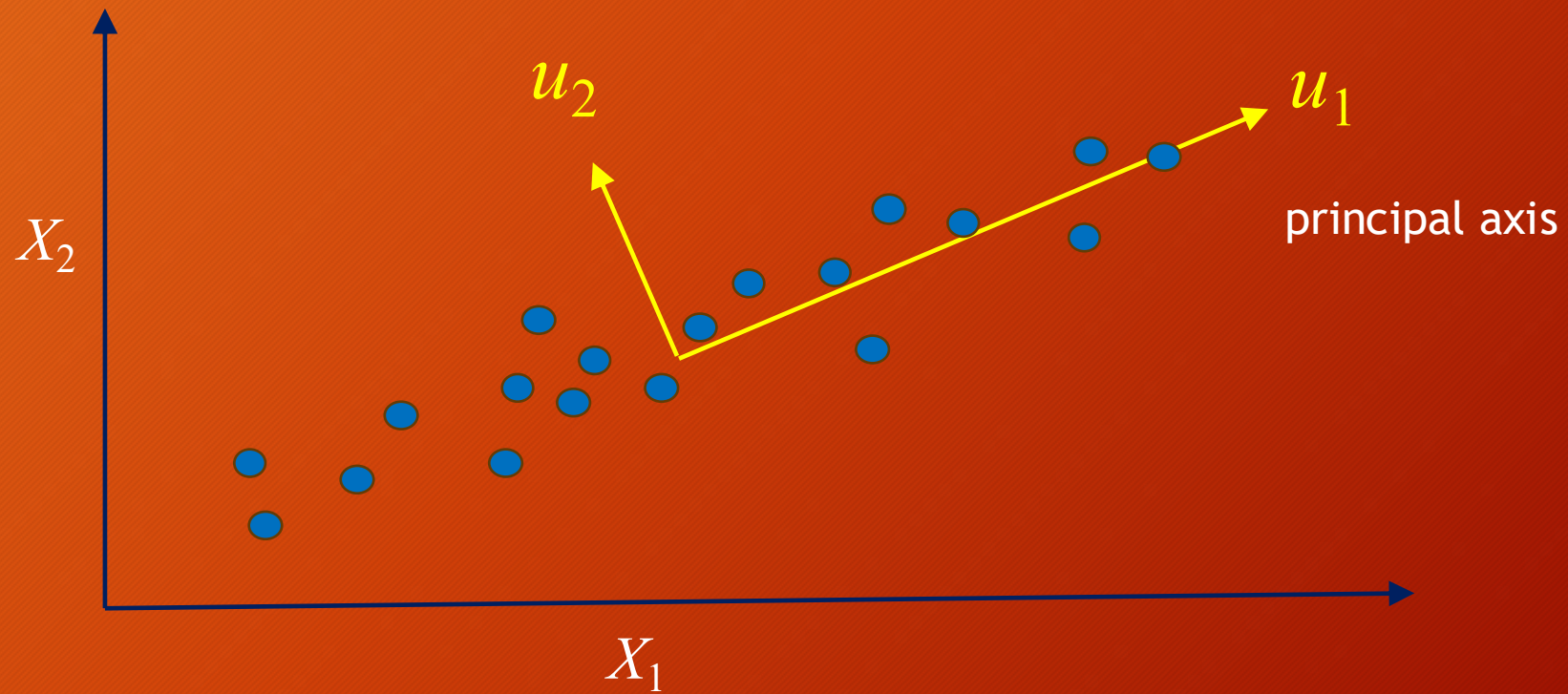
Convolutional Neural Networks (CNN)



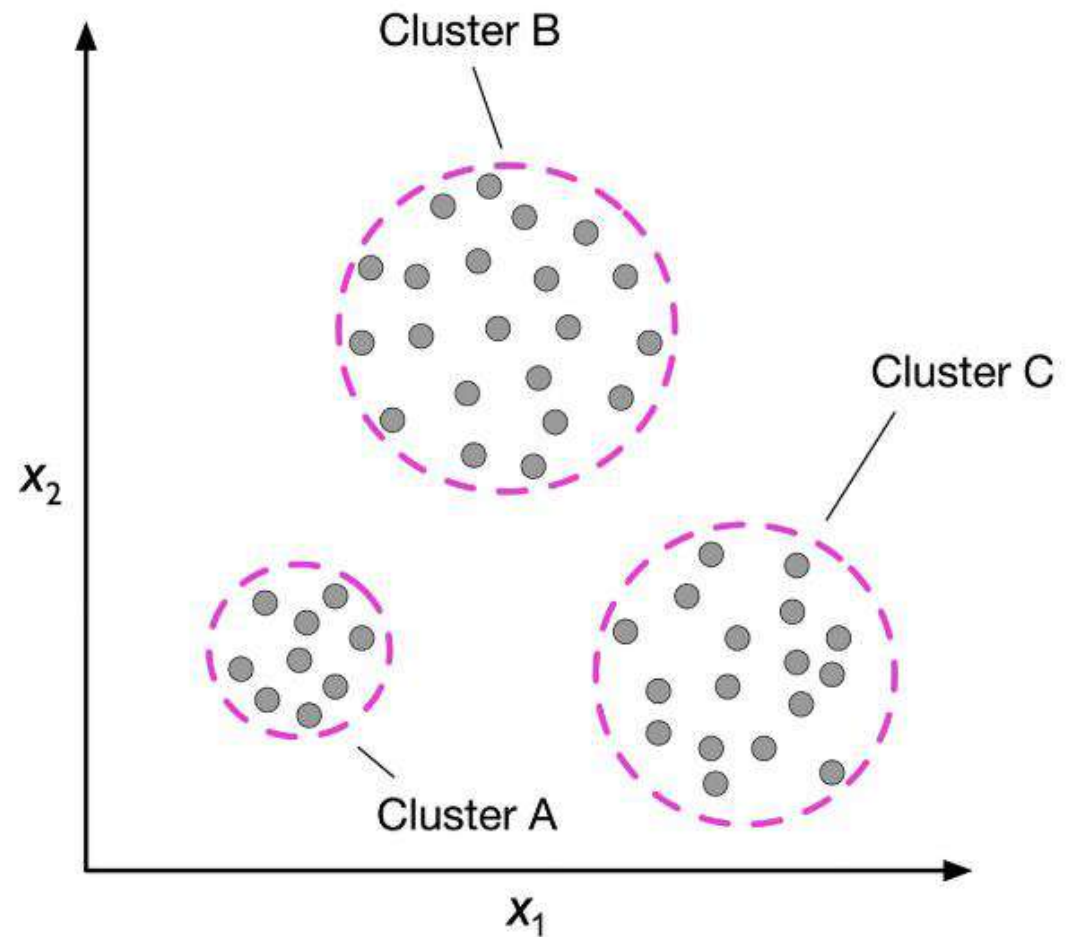
Unsupervised learning

Dimensionality Reduction

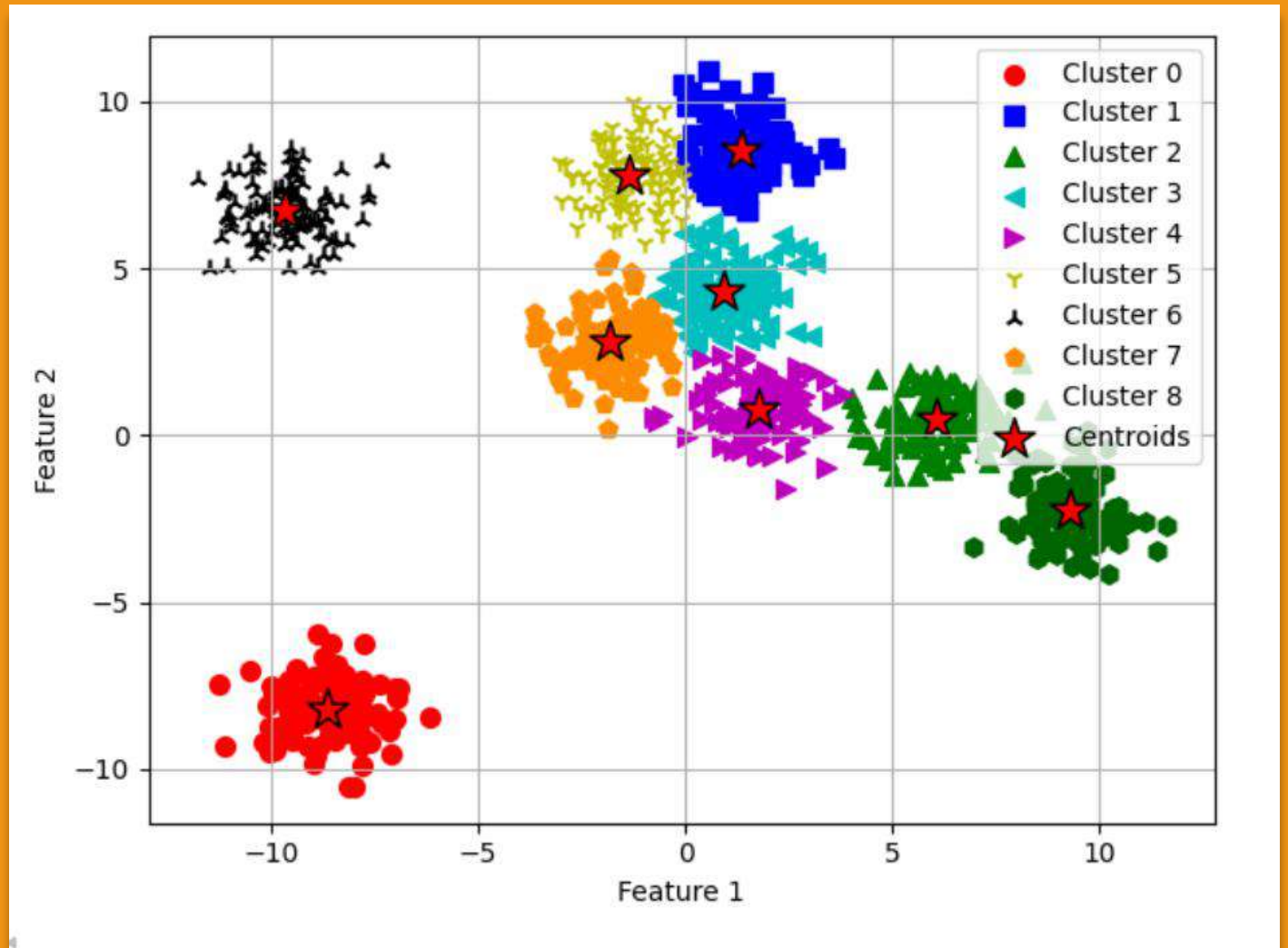
Principal Component Analysis (PCA)



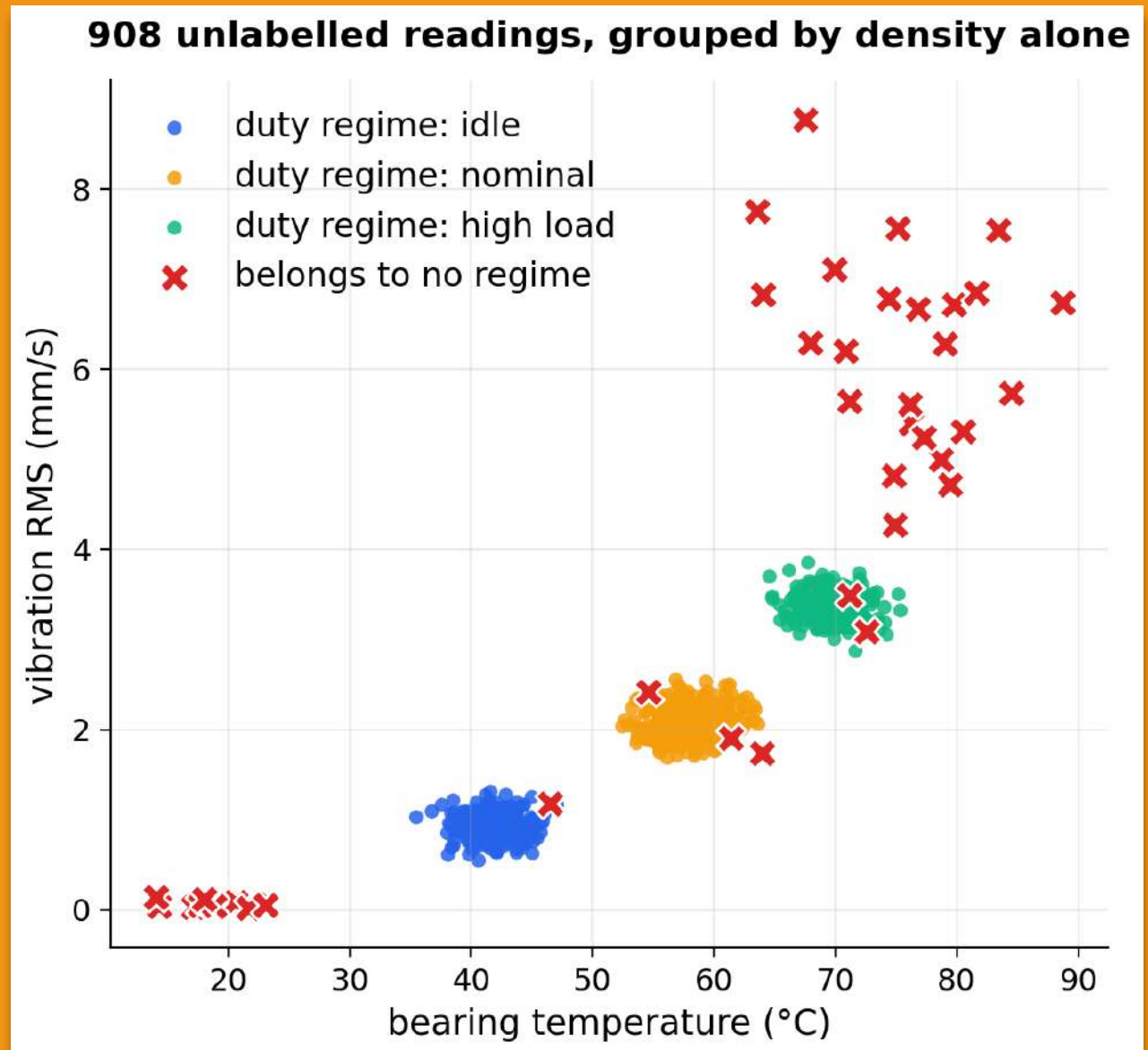
Unsupervised classification : clustering



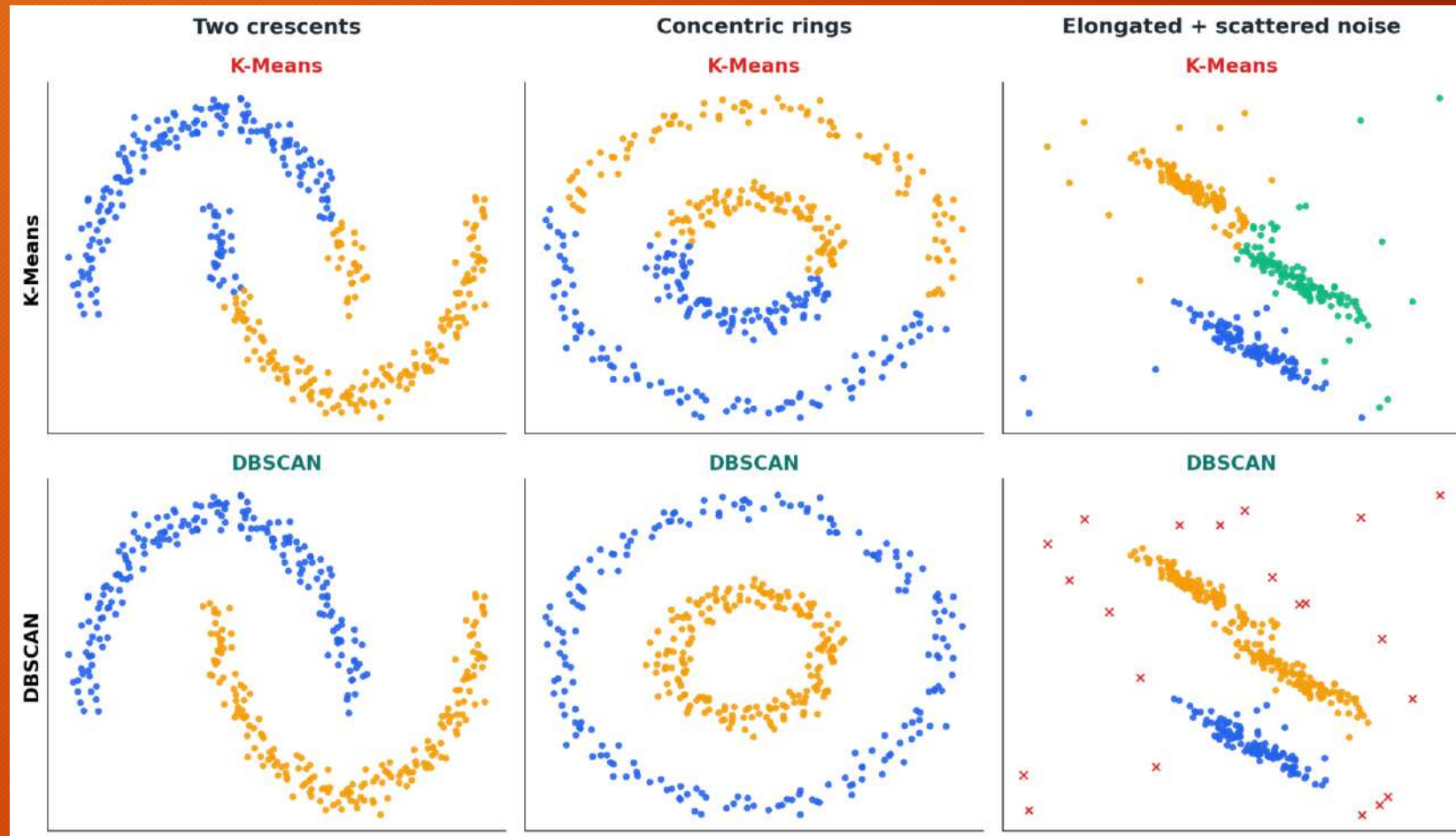
k-means algorithm



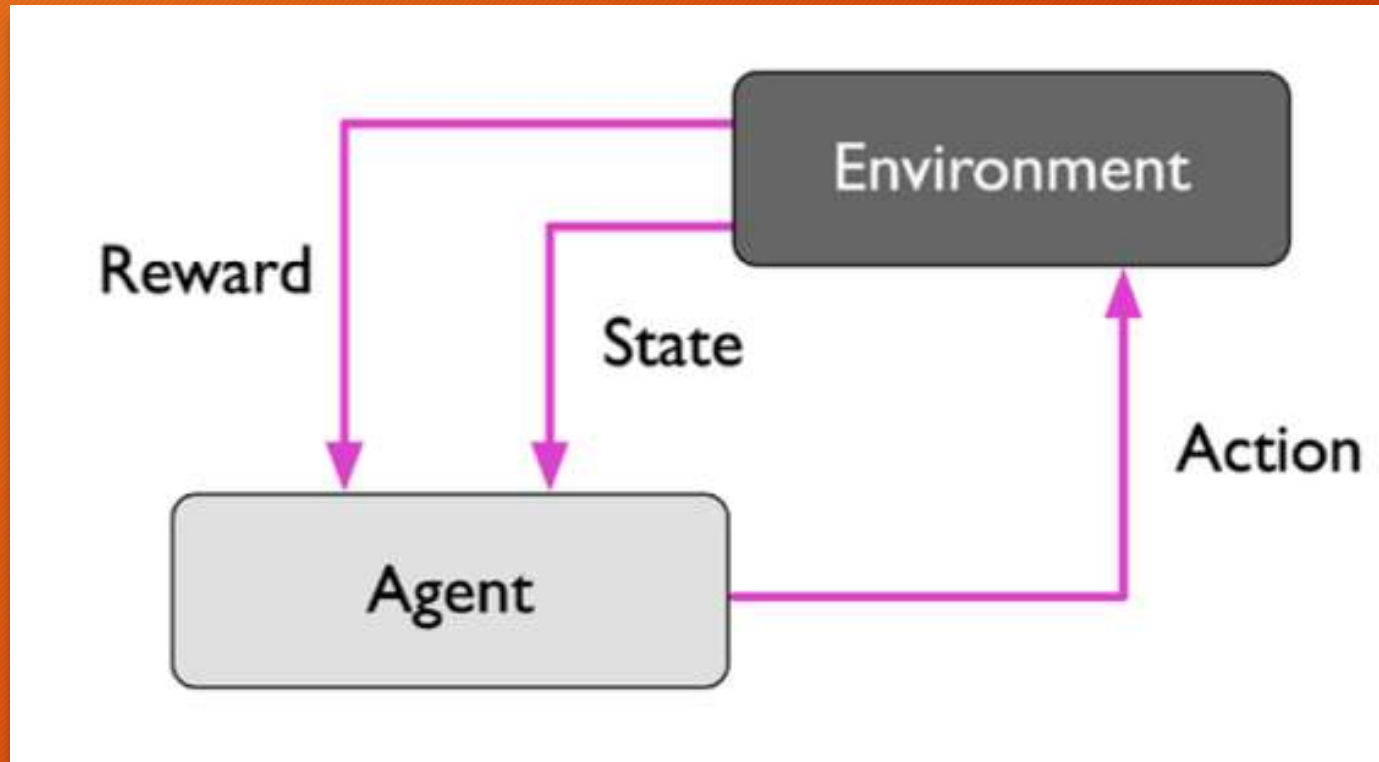
DBSCAN



k-means v.s. DBSCAN



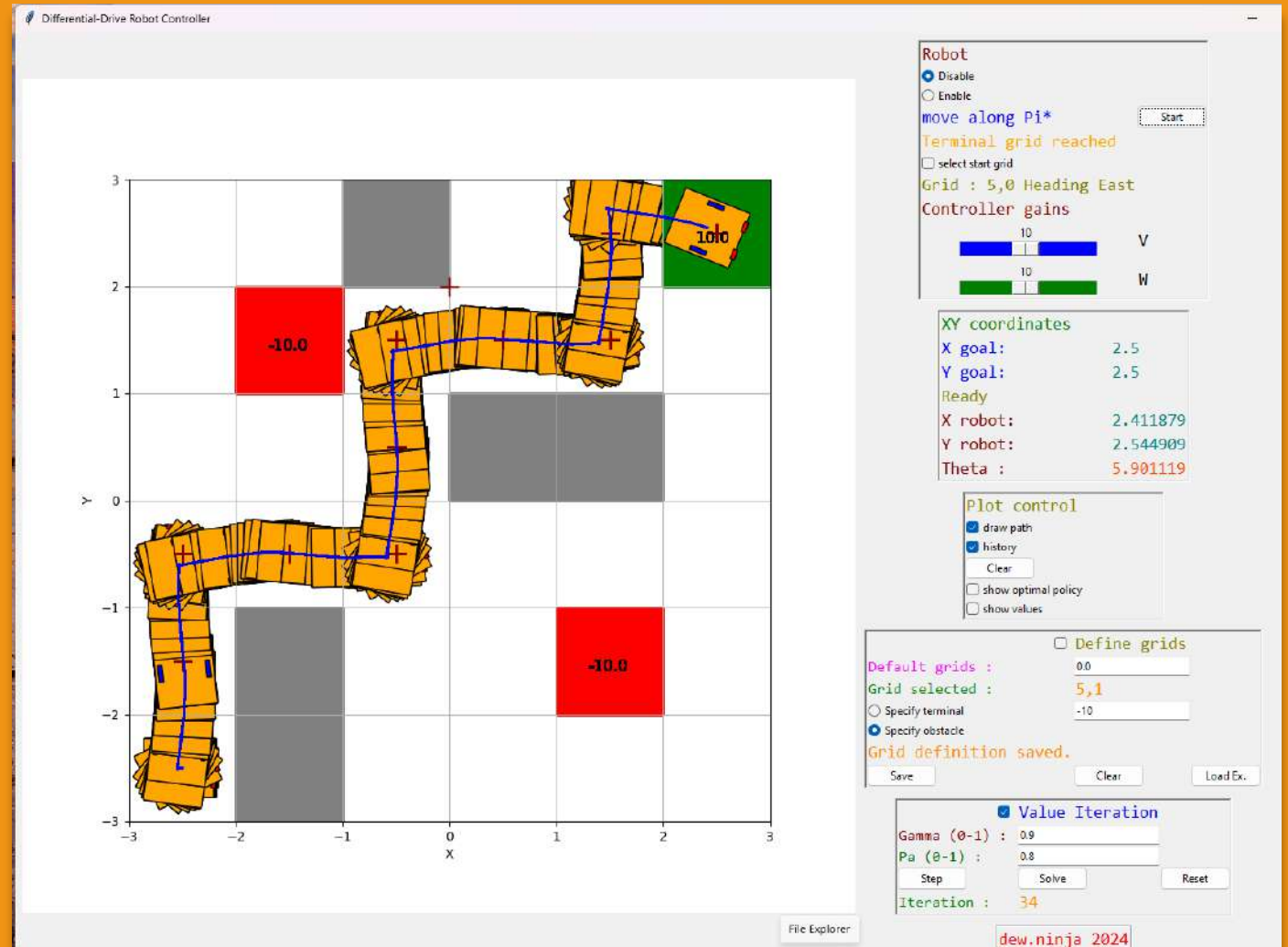
Reinforcement learning



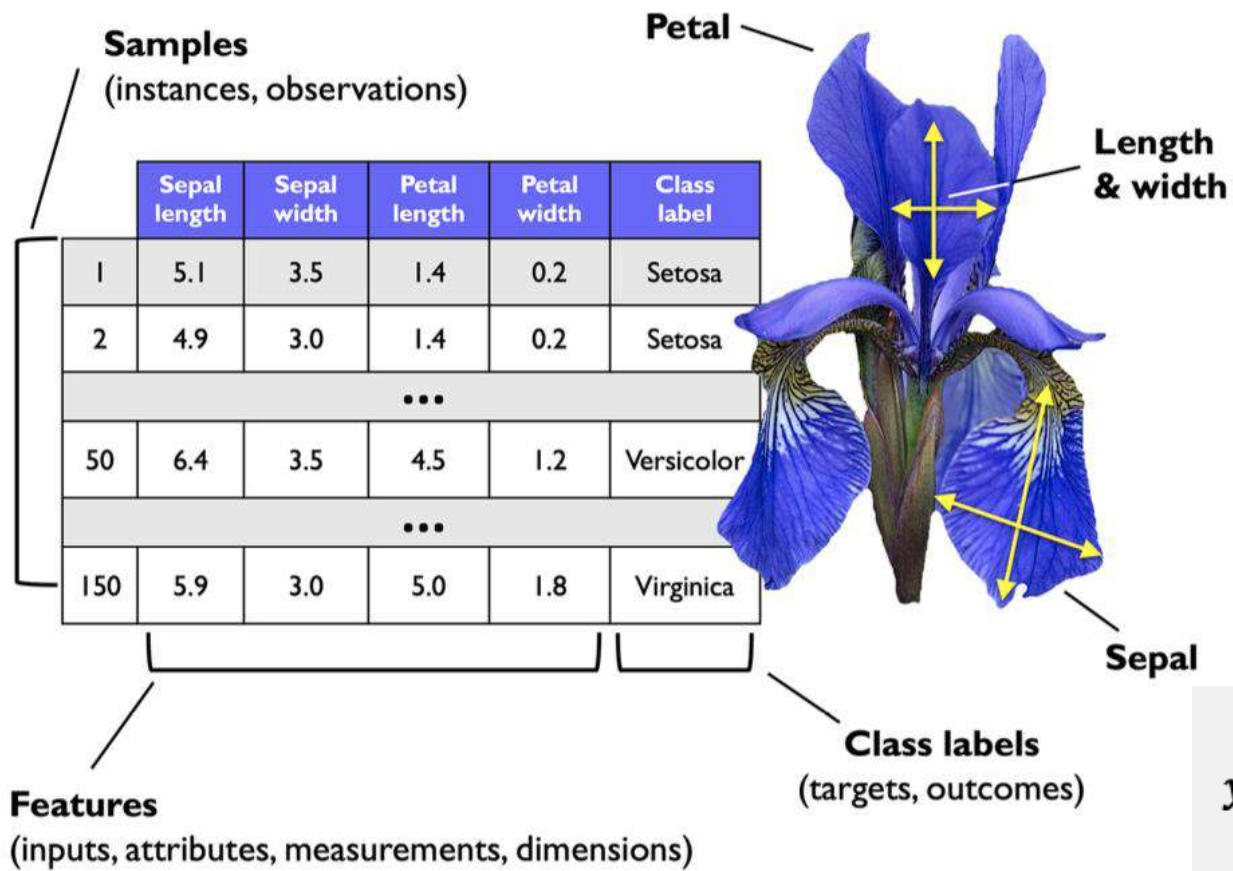
Reinforcement learning process

[1]

RL used in mobile robot path planning



Basic terminology and notations



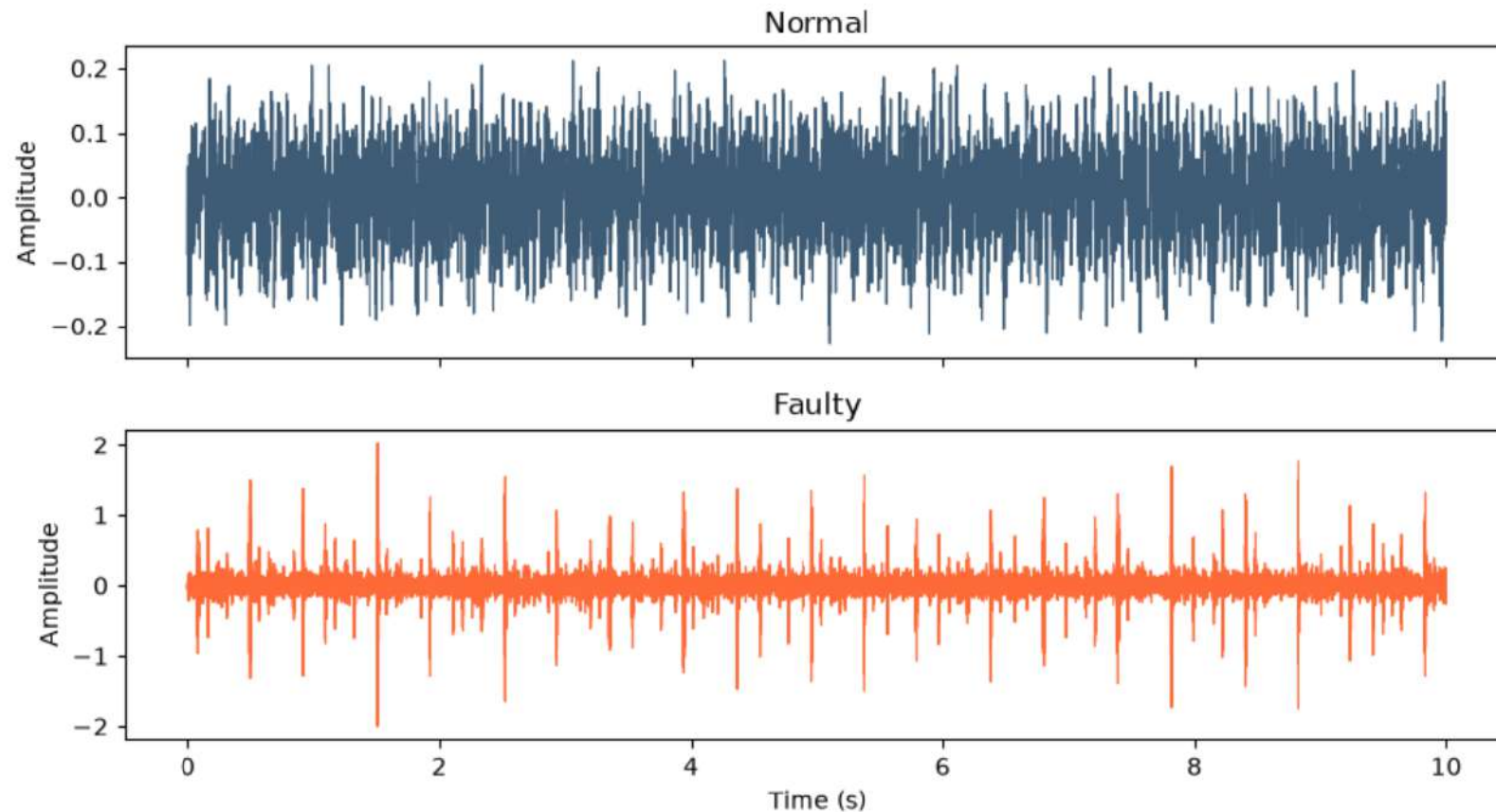
$$X \in \mathbb{R}^{150 \times 4}$$

$$\begin{bmatrix} x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & x_4^{(1)} \\ x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & x_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(150)} & x_2^{(150)} & x_3^{(150)} & x_4^{(150)} \end{bmatrix}$$

$$y = \begin{bmatrix} y^{(1)} \\ \dots \\ y^{(150)} \end{bmatrix}, \text{ where } y^{(i)} \in \{\text{Setosa, Versicolor, Virginica}\}$$

Example : the iris dataset

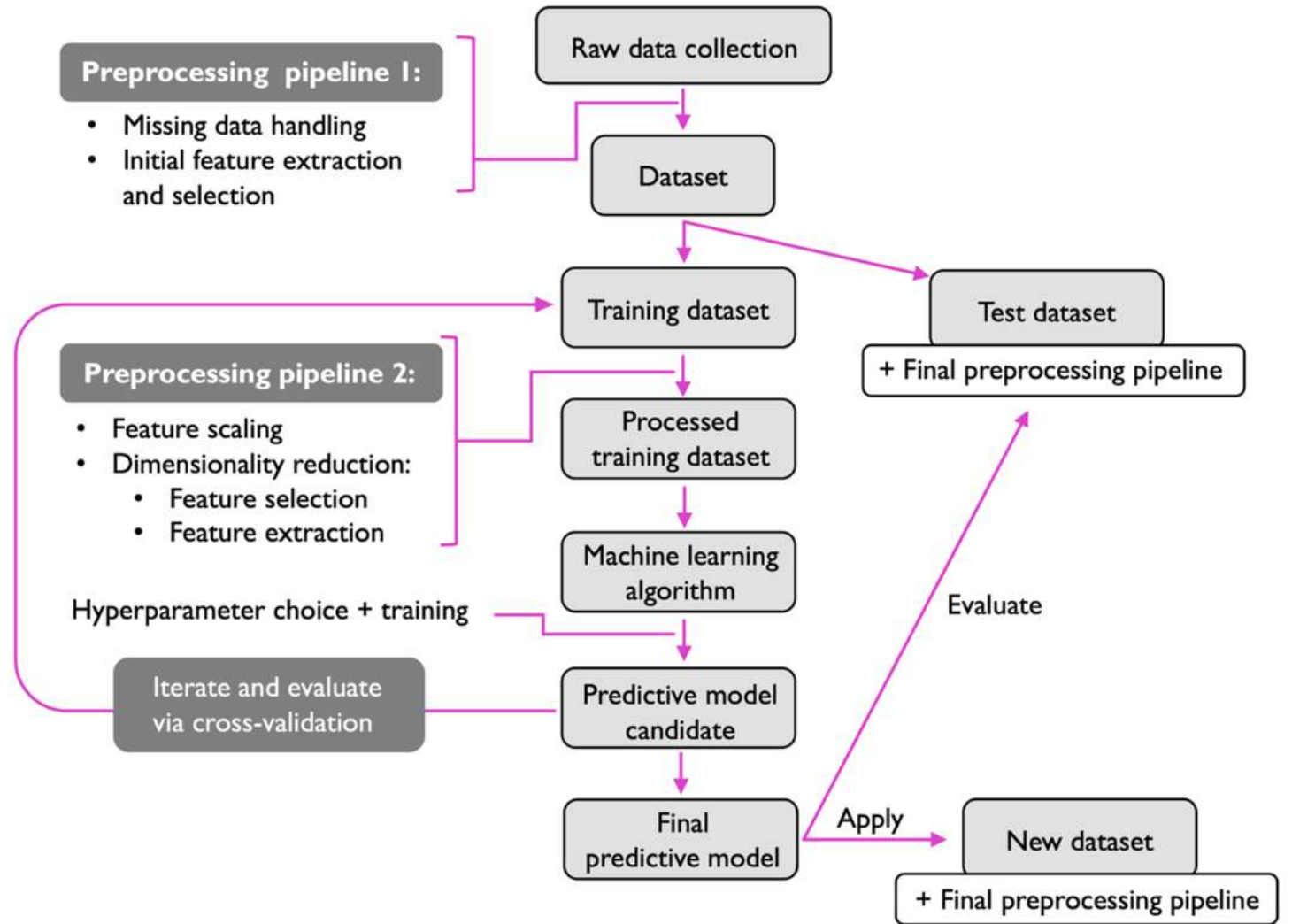
Example : Detect machine anomaly



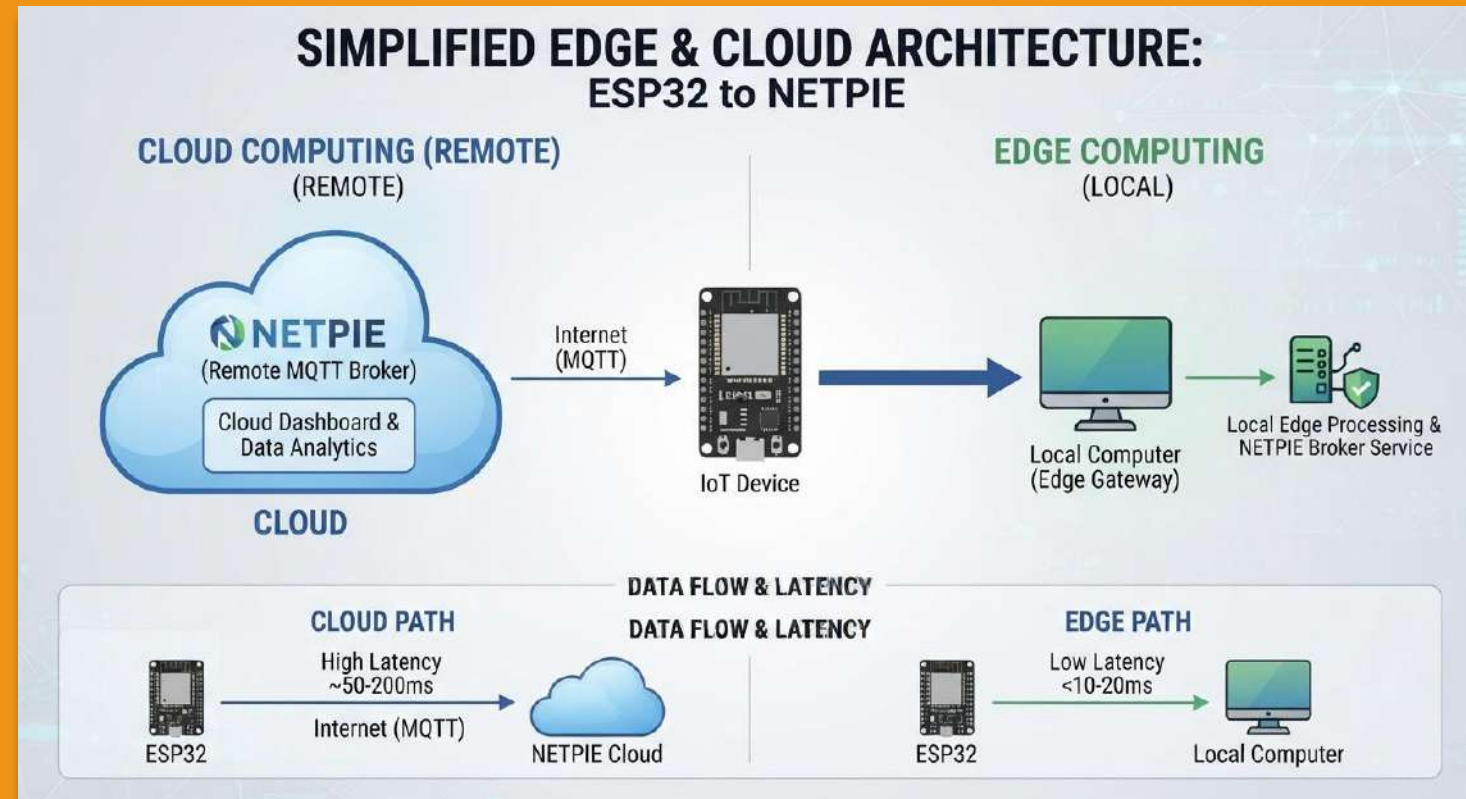
Extract features
X with label
0 = Normal
1 = Faulty

For example,
features could be
mean, std, rms, ptp

Building machine learning system



Edge v.s. Cloud Computing



Development

- Use Google Colab (has to install additional packages for every use)
- Create environment on your computer

Setup Python environment locally

Using conda

You should install anaconda or miniconda before executing the following commands on Anaconda prompt.

```
conda install -y jupyter
```

```
conda create -n mlpi
```

```
conda activate mlpi
```

```
conda install conda-forge::jupyterlab
```

```
pip install scikit-learn torch torchvision torchaudio
```

```
python -m ipykernel install --user --name mlpi --display-name "mlpi"
```

Then launch jupyter lab

Use uv

- Extremely fast Python package and project manager written in Rust
- More details at <https://docs.astral.sh/uv/>

Using docker

Install docker desktop

Create a work folder and put the provided config.yml and Dockerfile in there

run docker desktop

cd to that folder and type

- docker compose up

Open browser at local:8080

Specify password to launch jupyter notebook

config.yml

```
services:
  notebook:
    build: ./
    ports:
      - 8080:8888
    user: root
    environment:
      - DOCKER_DEFAULT_PLATFORM=linux/amd64
      - JUPYTER_TOKEN=yourpassword
      - NB_UID=1000
      - GRANT_SUDO="yes"
    volumes:
      - ./:/home/jovyan
```

Dockerfile

```
FROM --platform=linux/amd64 quay.io/jupyter/pytorch-notebook
```

```
USER root
```

```
RUN pip install --upgrade pip && \  
    pip install scikit-learn && \  
    fix-permissions "/home/${NB_USER}"
```

- Stanford CS229 (Andrew NG : Autumn 2018)
<https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShTMGgzqpfVfbU>
- Stanford CS229 (Spring 2022)
https://www.youtube.com/playlist?list=PLoROMvodv4rNyWOpJg_Yh4NSql4Z4vOYy
- MIT 6.036 (Fall 2020)
https://www.youtube.com/playlist?list=PLxC_ffO4q_rW0bqQB80_vcQB09HOA3ClV
- R. Sebastian Intro to ML (Fall 2020)
https://youtube.com/playlist?list=PLTKMiZHVd_2KyGirGEvKlniaWeLOHhUF3&si=CDL0ERiKTeZbK01U

Some useful
online
courses

Recommended text :

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.

<https://sebastianraschka.com/blog/2022/ml-pytorch-book.html>



References

1. S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
2. F. Chollet. Deep Learning with Python 2ed. Manning Publications Co. 2021.
3. A. Zhang, Z. Lipton, M. Li and J. Smola. Dive into Deep Learning. 2022.
<https://d2l.ai/>
4. A. Ng. CS229 Lecture Notes. Stanford University. 2023.
https://cs229.stanford.edu/main_notes.pdf
5. T. Broderick. Introduction to Machine Learning course (6.036). Massachusetts Institute of Technology. 2020.
<https://tamarabroderick.com/ml.html>